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Disaster Management

Disaster

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- Disaster is a sudden, calamitous event bringing great damage, loss, destruction, and devastation to life and property.
- The damage caused by disasters is immeasurable and varies with the geographical location, climate, and the type of the earth's surface/degree of vulnerability.
- This influences the mental, socio-economic, political, and cultural state of the affected area.

Two Types **(depends upon their potential to cause damage)**

Major disasters: Hurricanes, Floods / Sea Surges / Tsunamis Snow storms, Earthquakes, Landslides,

Minor disaster: Localized and intensity of damage is much less

Ex., Landslides, fire accidents

Effect of Major Disasters

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- **Primary and Secondary**
- Deaths, Severe injuries, requiring extensive treatments
- Increased risk of communicable diseases, Damage to the health facilities
- Damage to the water systems
- Food shortage and nutrition demand
- Population movements, Social reactions
- Climatic exposure
- Mental health
- Damage to health infrastructure

India's Natural Disasters Proneness

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On the basis of geographic and climatic considerations, India can be divided into 5 Zones according to its disaster proneness to natural disasters

1. **Northern mountain region, including foothills:** Snow storms, Landslides, Cold waves, earthquakes.
2. **Indo-Gangetic plains**: heavy rains during monsoon make these plains vulnerable to Floods.
3. **Deccan plateau:** a drought-prone area.
4. **The western desert** is a drought-prone area.
5. **Coastal areas** are prone to Sea erosion, Cyclones, and Tidal waves

Disaster Management

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- Overall measures to minimize loss and recover from hazard impact
- Disaster management is the discipline that involves **preparing, warning, supporting, and rebuilding societies when natural or man-made disasters occur.**
- It is a **sequential and continuous process** by which all individuals, groups, and communities manage hazards in an effort to avoid or minimize the impact of disasters resulting from hazards.

<u>Disaster Cycle Consists of two phases</u>

1. Risk Reduction Phase, before a Disaster
2. Recovery Phase, after a Disaster

Disaster Management aspects

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- **Prevention**
- **Preparedness:** Plans or preparations made to save lives or property, and help the response and rescue service operations. Eg: early warning systems and capacity building
- **Response:** includes actions taken to save lives and prevent property damage, and to preserve the environment **during** emergencies or disasters. Eg: emergency evacuation, quarantine.
- **Mitigation:** To prevent hazards from developing into disasters altogether, or to reduce the effects of disasters when they occur.
- **Risk Reduction:** anticipatory measures and actions that seek to avoid future risks as a result of a disaster.
- **Rehabilitation or Recovery:** includes actions that assist a community to return to a sense of normalcy after a disaster.
- **Reconstruction**

Mitigative measures can be structural or non-structural.

Structural: technological solutions flood levees, community grain stores, flood reservoir in drought.

Non-structural: include legislation, land-use planning (eg. the designation of nonessential land like parks to be used as flood zones), and insurance.

Preparedness measures include the Risk assessment:

1. Hazard mapping: Reveals areas which susceptible to hazards
2. Vulnerability analysis and casualty prediction: Asses the vulnerability of person or property to the hazard which gives planners an idea of what resources need to be in place to respond to a particular kind of event.
3. Resource assessment: Assessment of resources needed top improve disaster planning and development such as ··stockpiling, inventory, and maintenance of supplies and equipment
4. Exercise

POST DISASTER PREVENTIVE MEASURES

- Maintenance of law and order, prevention of trespassing, Evacuation.
- Recovery of dead bodies and their disposal and medical care for the injured.
- Supply of food and drinking water and temporary shelters like tents, metal sheds etc.
- Repairing lines of communication and information, Restoring transport routes.
- Quick assessment of destruction and demarcation of destroyed areas
- Cordoning off severely damaged structures that are liable to collapse during aftershocks.

Recovery

- *It differs from the response phase in its focus; recovery efforts are concerned with issues and decisions that must be made after immediate needs are addressed*.
- Recovery efforts are primarily concerned with actions that involve rebuilding destroyed property, re-employment, repair of other essential infrastructure, compensation for loss of life and property.
- **Window of opportunity'** for the implementation of mitigative measures that might otherwise be unpopular.

Disaster Management: Indian Scenario

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National Policy on disaster management Involves (National Institute of Disaster Management NIDM)

It aims at:

- **prevention, mitigation** and **preparedness** in **pre-disaster phase**
- **Creation of awareness** for disaster reduction and for **improving**
- **preparedness amongst the communities,** using media, school education, and the network of the building centre.
- **Appropriate amendments** in the legislative and regulatory instruments
- **assessment surveys and investigations of the nature and extent of damage in post disaster situations.**

FLOODS

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- Overflow of water causing an expanse of water-submerged land

Main causes:

1. Heavy rainfall in low lying area and areas without proper drainage
2. Melting of ice
3. Coastal flood-Tsunami: a series of water waves (called a tsunami wave train) caused by the displacement of a large volume of a body of water, usually an ocean

- **Types of Floods**: Flash flood, River flood, Coastal flood

- *Chennai flood, 2016*
- *Uttarakhand flood in 2013*

North Indian Flood in 2023

Flooding in the Netherlands and how are they tackling

Consequences of floods

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Primary effects

Physical damage and Casualties

Secondary effects:

1. **Water supplies**- Contamination of water. Clean drinking water becomes scarce.
2. **Diseases**
3. **Crops and food supplies**- Shortage of food crops can be caused due to loss of entire harvest.
4. **Economic**
5. **Soil Erosion and Loss of Soil fertility**

Flood Disaster Management

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Structural:

- Dams and Reservoirs
- Embankments, flood walls, sea wall
- Natural detention basin
- Channel improvement
- Drainage improvement
- Diversion of floodwaters.
- Protection by Wetlands

Non-structural measures

- Flood plain management
- Flood proofing including disaster preparedness, and response planning and Flood forecasting and Warning
- Disaster Relief
- Flood fighting including Public Health Measures

DROUGHT

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- extended period where water availability falls below the statistical requirements for a region

Consequences of drought:

Desertification, Famine, Social and economic drop

Droughts can be of three kinds:-

(i) **Meteorological drought**: Actual rainfall in an area is significantly less than the climatological mean of that area.

(ii) **Hydrological drought:** Drying water reservoirs. A marked depletion of surface water causing very low stream flow and drying of lakes, rivers and reservoirs

(iii) **Agricultural drought:** Inadequate soil moisture resulting in acute crop stress and fall in agricultural productivity

Drought management:

- Water conservation, Dams, and lakes construction
- Using alternative technology for irrigation - drip irrigation
- Networking of rivers
- Social forestry method: the Kolar district of Karnataka is social forestry under world bank aid.
- Drought by Eucalyptus plantation- due to the high transpiration

EARTHQUAKE

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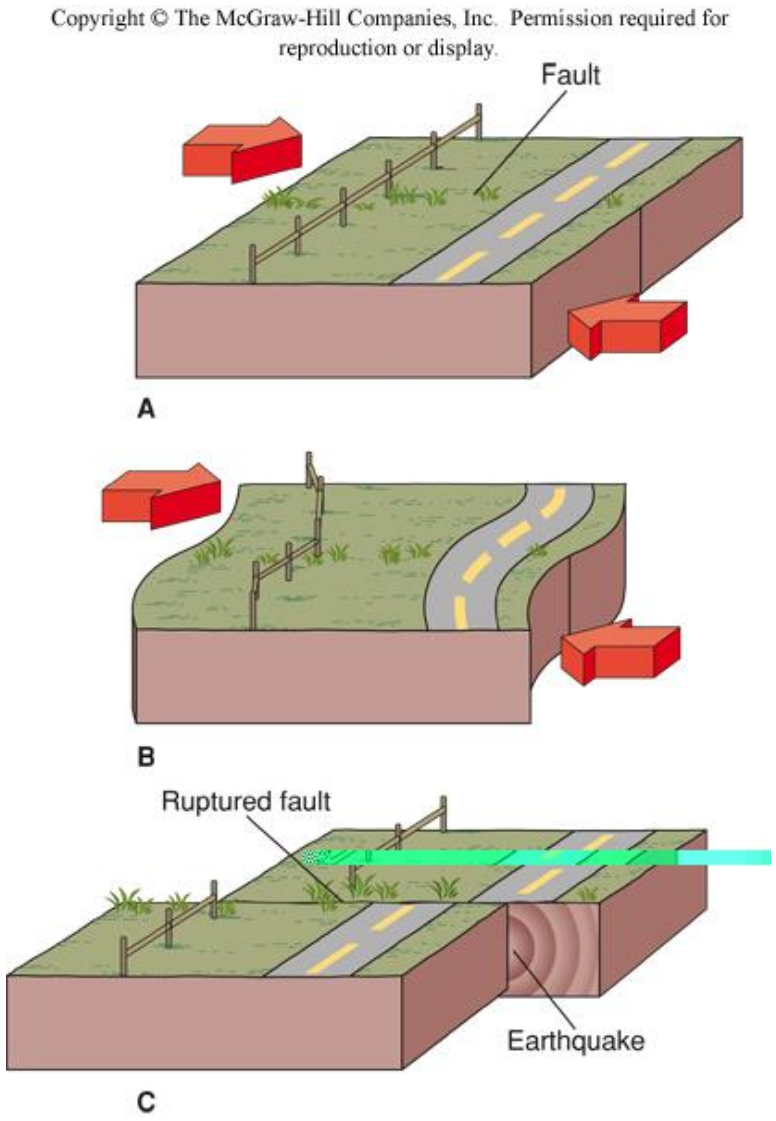
- An earthquake is a sudden motion or trembling of the ground produced by the abrupt displacement of rock masses along the faults.

Energy is released during earthquakes in the form of *seismic waves.*

Elastic rebound theory

- The sudden movement within the crust or mantle create concentric waves from the point.
- The point where the earthquake's motion starts is called focus
- **The epicenter** is the point on the earth's surface that is directly above the focus.
- **Main Causes:** Tectonic activity, Volcanic Action, Man made

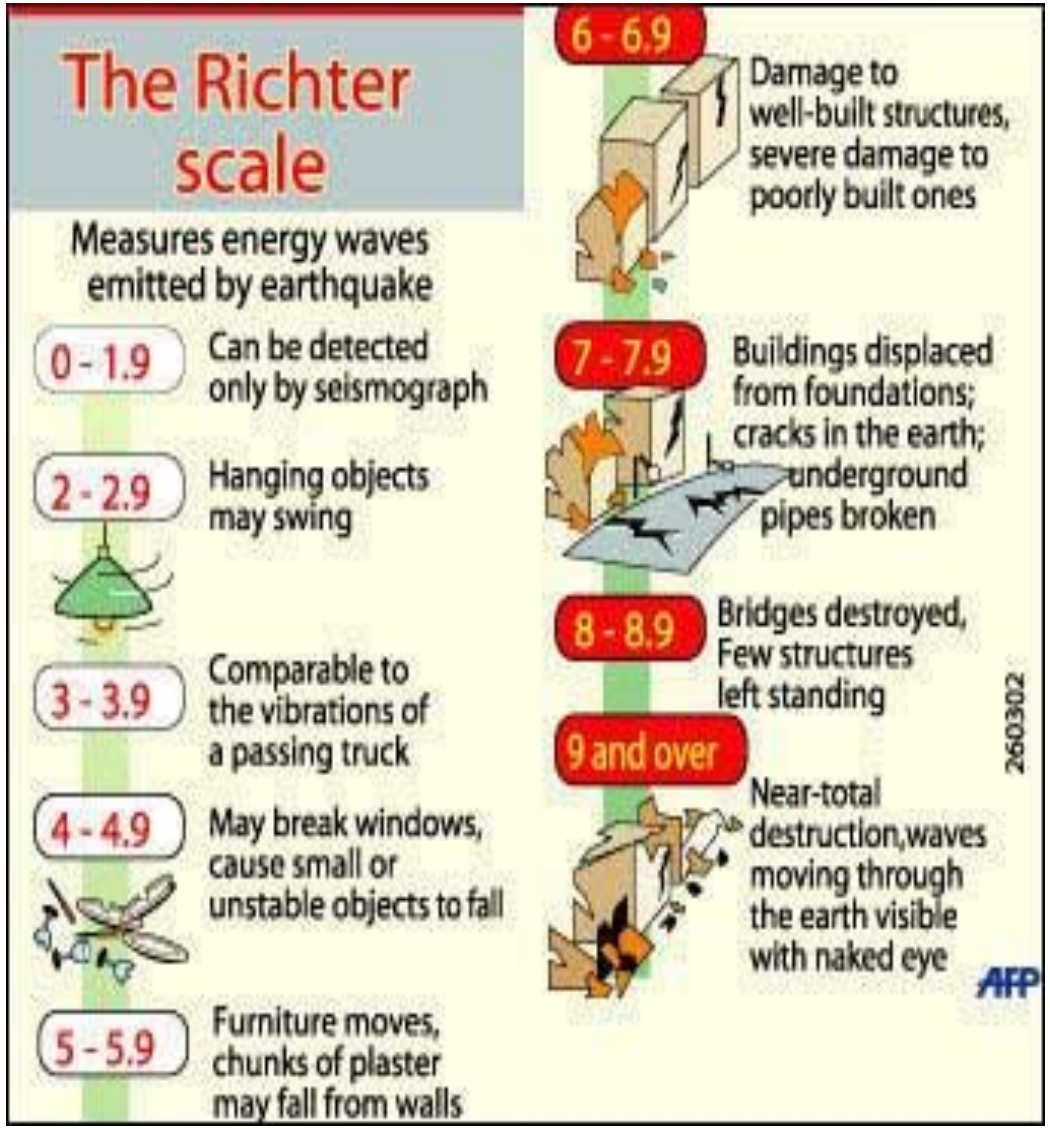


- Shallow earthquakes (90%) = < 70 km
- Intermediate earthquakes = 70 to 300 km.
- Deep earthquakes (3%) = >300 km deep.

HOW IS EARTHQUAKE INTENSITY MEASURED

1) Mercalli Intensity scale (qualitatively based)

2) Richter Magnitude scale (quantitatively based)



Factors Effecting Destructiveness of Earthquake

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- **Magnitude:** The more energy released, the greater potential for damage, regardless of all other factors (& greater areal extent).
- **Distance:** The more energy released, the greater potential for damage, regardless of all other factors (& greater areal extent).
- **Local soils & bedrock geology:** Soil characteristics may amplify shaking. Complex geology tends to dampen waves
- **Building codes & materials**

SECONDARY EFFECTS: It is this phenomena that actually kill people, i.e., ground-shaking does not usually harm an individual

Ground shaking (structural collapse), Fires (in urban settings; broken gas lines), Landslides, Ground liquefaction, Seismic sea waves (Tsunami's)

Caused by SURFACE waves (arrive last)

Earthquake Disaster Management

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PRE-DISASTER PREVENTIVE

Structural:

- Building code
- Constructing earthquake-resistant community buildings and buildings
- Retrofitting of weak structures in highly seismic zones.

Non-structural

- Re-framing buildings' codes, guidelines, manuals and bye laws and their strict implementation. Tougher legislation for highly seismic areas.
- Evolving educational curricula

Preparedness:

- Response doctrines for state and local government
- Preparedness plans for individuals and businesses.
- Train communities in high-risk areas in post-disaster search, rescue and relief.
- Practice an extensive programme of mass drills in high-risk areas for earthquake damage reduction.
- Train local NGOs and strengthen their capacity and capabilities.
- Inculcate basic know-how amongst school kids and train field personnel.

Earth quake response:

- Evacuating people to safer pre-designated areas
- Emergency sanitation, alternative supplies of water, custody of valuables,
- Provision of welfare facilities
- Removal of debris
- Disposal of dead and their identification
- Restoration of communications
- Mobilization of transport.

Recovery or Post-disaster

- Detailed survey for assessment of damage and repair/ reconstruction and seismic strengthening or demolition.
- Selection of sites for new settlements.
- Review of the existing seismic zoning maps and risk maps.
- Review of seismic codes and norms of construction.
- Training of personnel, engineers, architects, builders and masons.

Camp-work and employment Outside relief.

COASTAL HAZARDS

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- **Coastlines are highly dynamic systems and are very prone to hazards and disasters.**

Major types of coastal hazards are:

- **Coastal inundation:** refers to flooding of dry, low lying coastal areas, mainly caused due to severe weather changes along coast
  - Storm surges
  - Tsunamis
  - Sea level rise
- **Coastal erosion**

Storm surges

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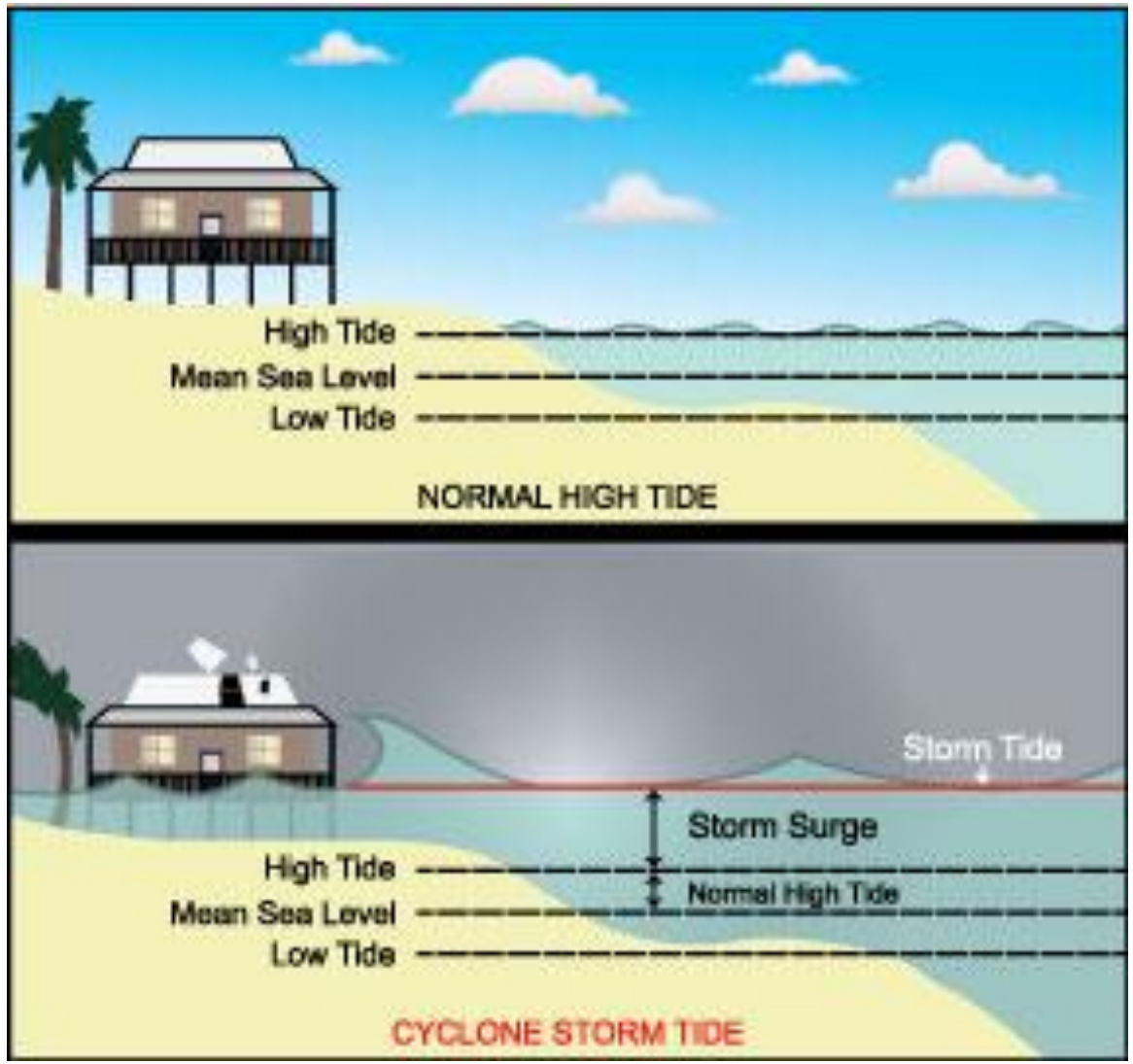
- Abnormal rise in water generated by a storm above and over the astronomical tide.
- Its mere a difference between water levels, no reference level present.
- Caused primarily by strong winds in hurricanes or cyclones.
- Storm surge increases water level along with high tides, waves, freshwater inputs.



Cyclones/hurricanes

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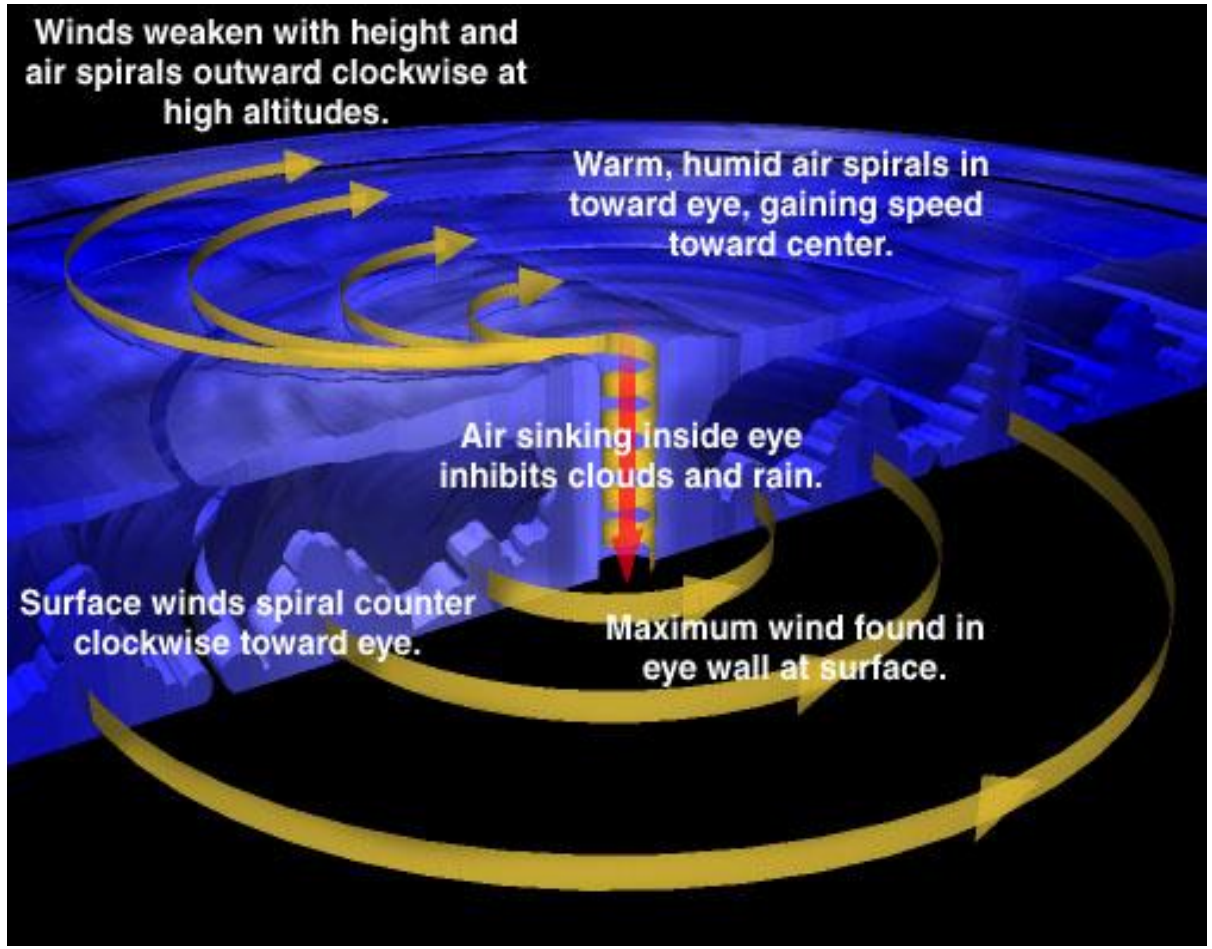
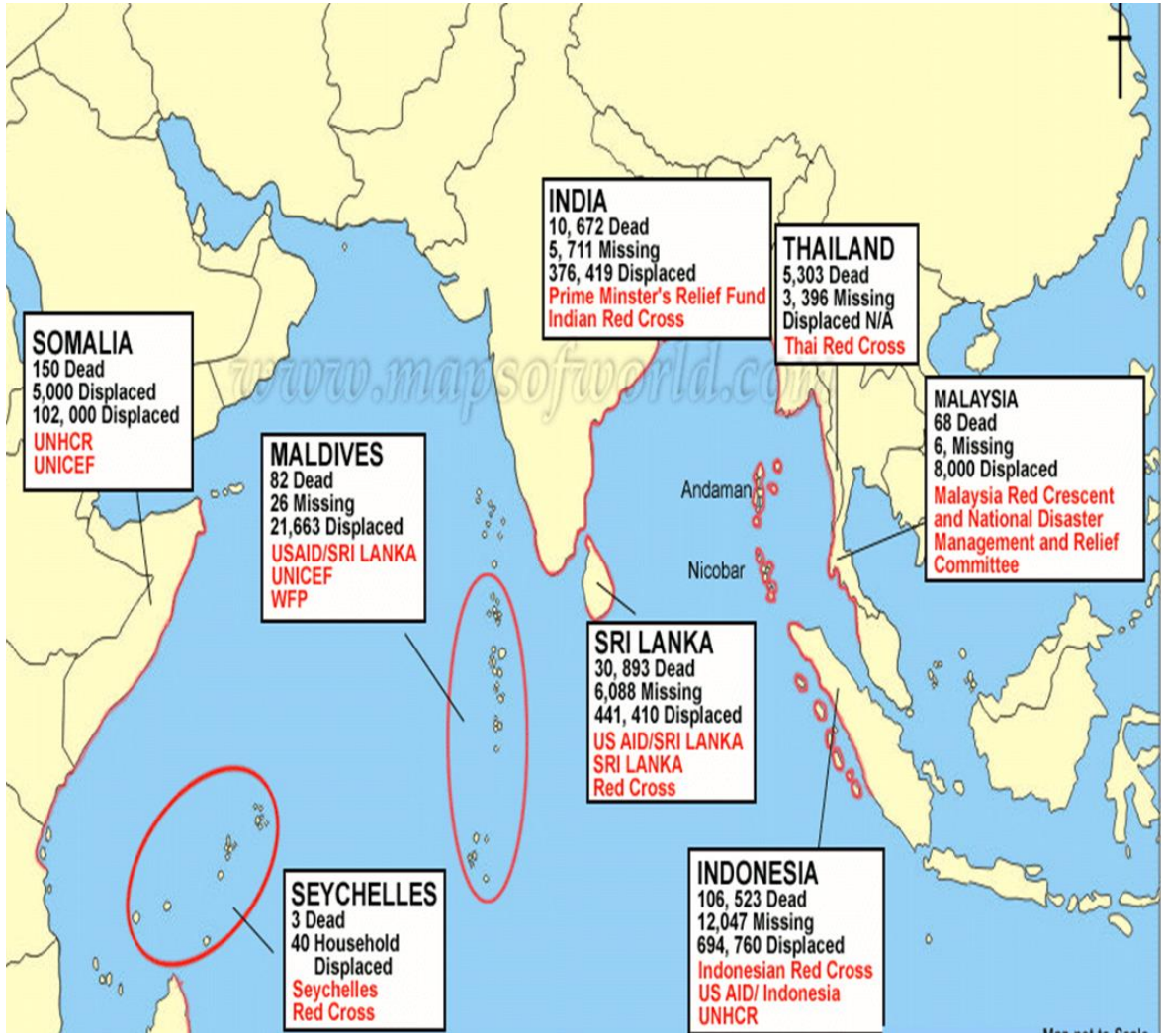
- Tropical cyclones form only over warm ocean waters near the equator, warm core cyclones.
- Cyclones are large scale air mass which revolves around strong centers of low pressure.
- Wind spirals CC in northern hemisphere and Clockwise in southern hemisphere.
- Hurricanes is the name given to cyclones in Atlantic and Northeast pacific, cyclone is name in South Pacific and Indian Ocean whereas in Northwest Pacific, they are known as Typhoons.



Damages due to cyclone is caused: Wind speed, Storm surges, Torrential rain and Flooding

Tsunamis

- Tsu (harbor) and name(waves) is a Japanese word.
- Are long wavelength, shallow-water, progressive waves.
- These are caused by underwater earthquakes (sudden vertical movement of Earth faults), landslides, volcanoes etc.
- Properly called as seismic sea waves.
- When these reaches shore, velocity drops, time period remains constant, this leads to high increase in wavelength.
- Water surges offshore as high tides with immense force.



SEA LEVEL RISE

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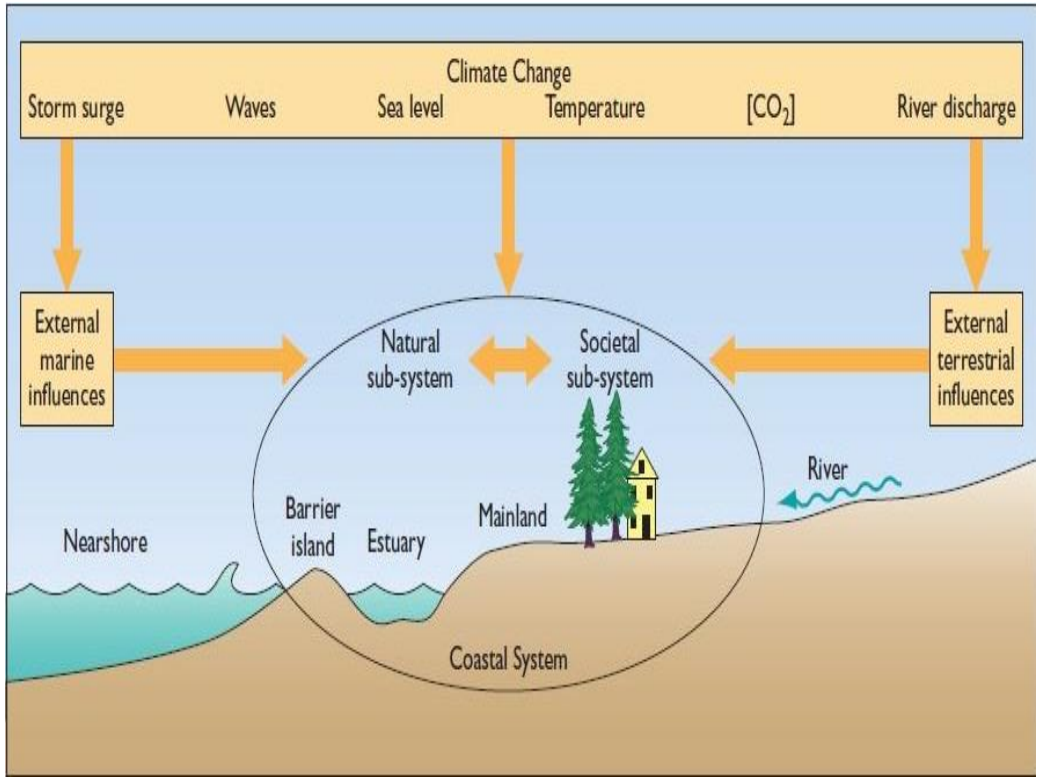
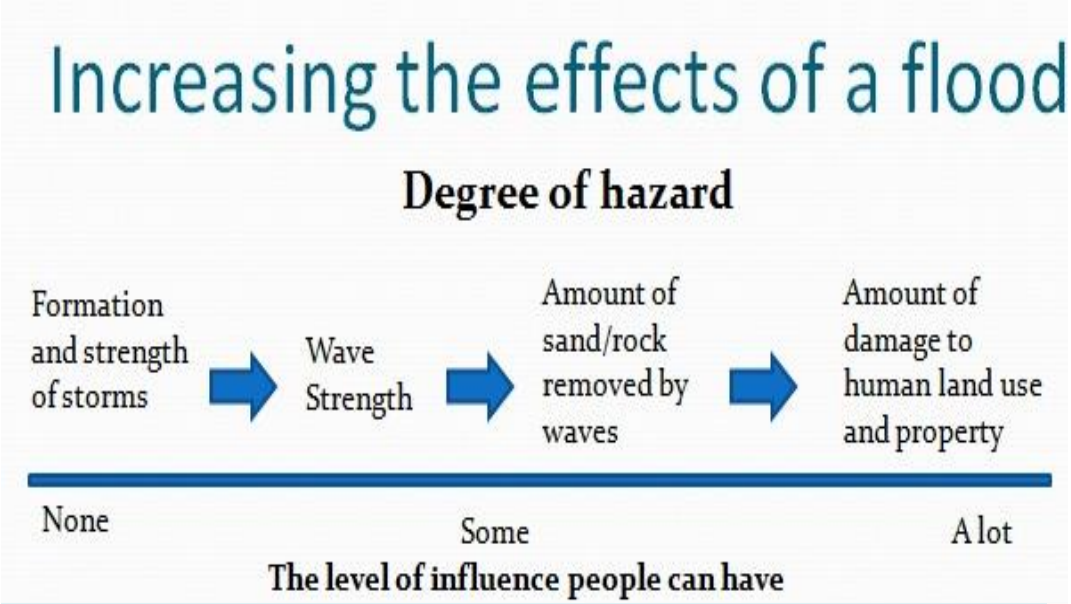
- Coastal deltas plains are subsiding naturally under their heavy sediment load,

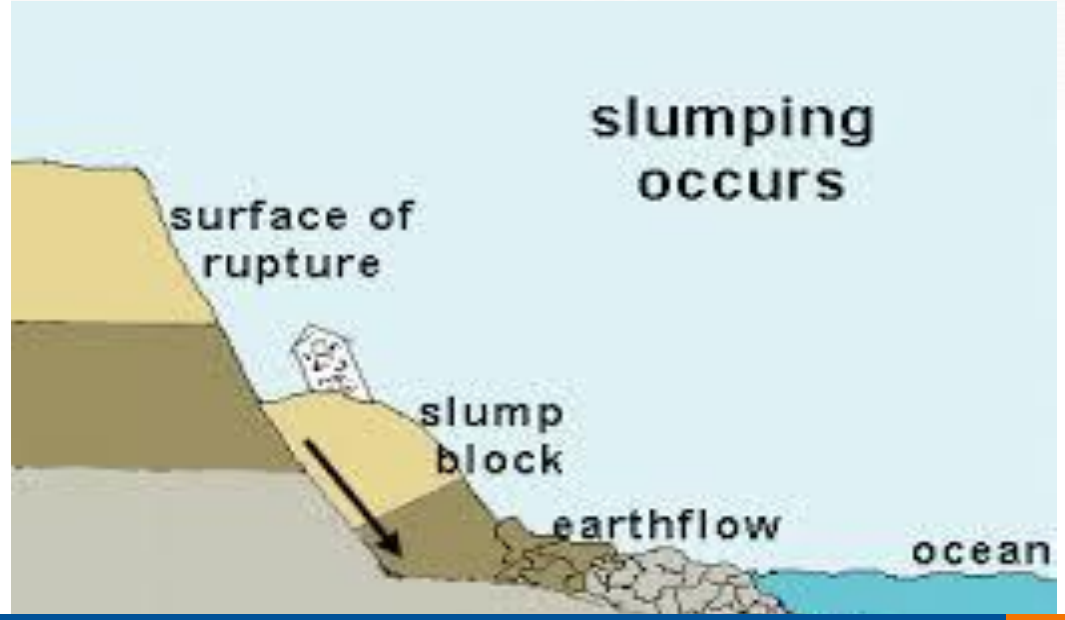
This accelerates the relative rise of sea level making it greater than the global average.

- The intrusion of seawater into groundwater aquifers will contaminate the freshwater supplies of many coastal communities.
- Submergence of some inhabited islands in the Pacific Ocean and the Maldives, a chain of coral atolls in the Indian Ocean



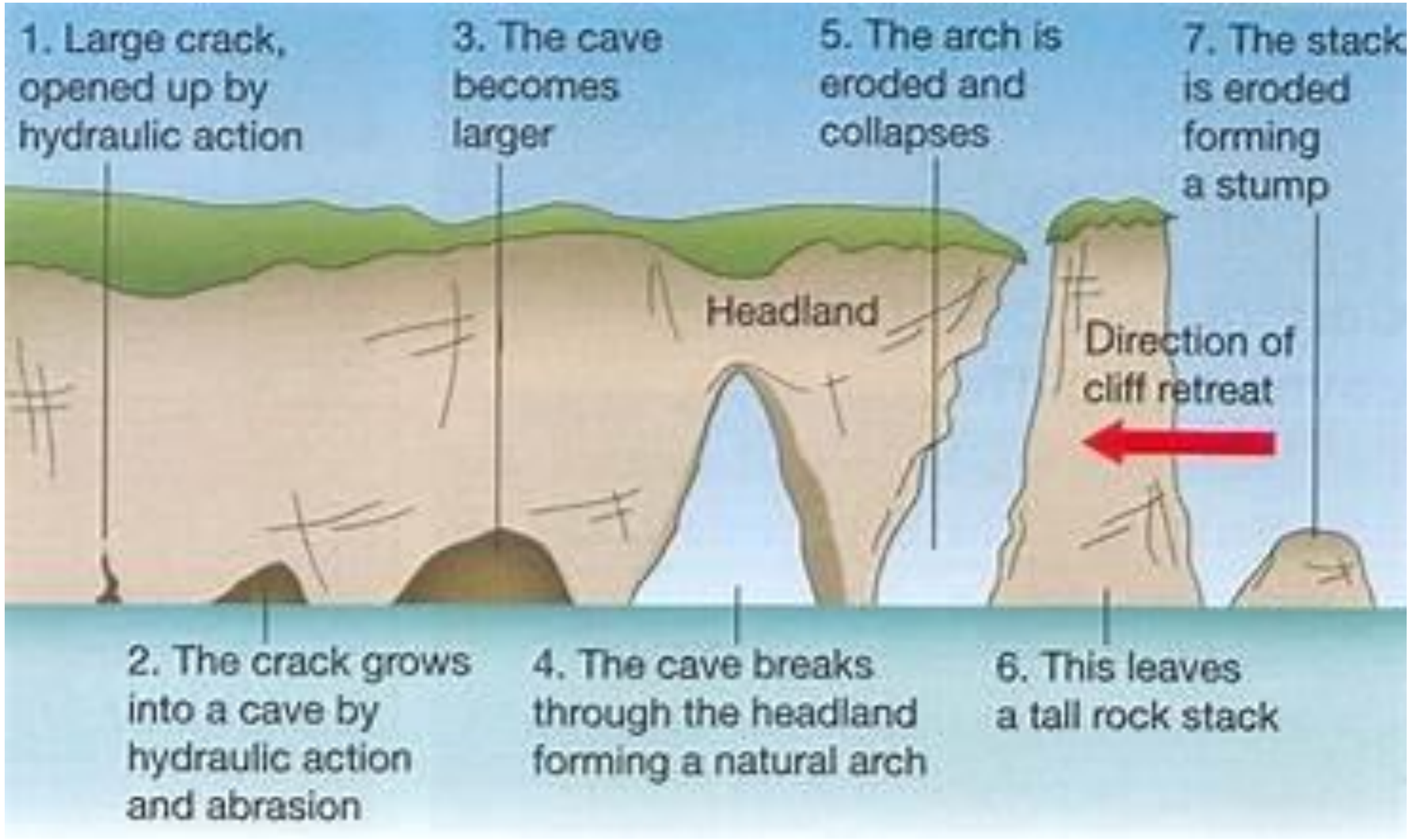
COASTAL EROSION





- Wearing away of land and beaches, dunes near coasts.
- Three major ways:
  - **Hydraulic (includes breaking and loosening)**
  - **Abrasion (includes rubbing and grinding)**
  - **Corrosion**

Most rapid on high energy coast. Erosion is concentrated near average sea level.


1. Large crack, opened up by hydraulic action
3. The cave becomes larger
5. The arch is eroded and collapses
7. The stack is eroded forming a stump
Headland
Direction of cliff retreat
2. The crack grows into a cave by hydraulic action and abrasion
4. The cave breaks through the headland forming a natural arch
6. This leaves a tall rock stack

COASTAL ADAPTATION

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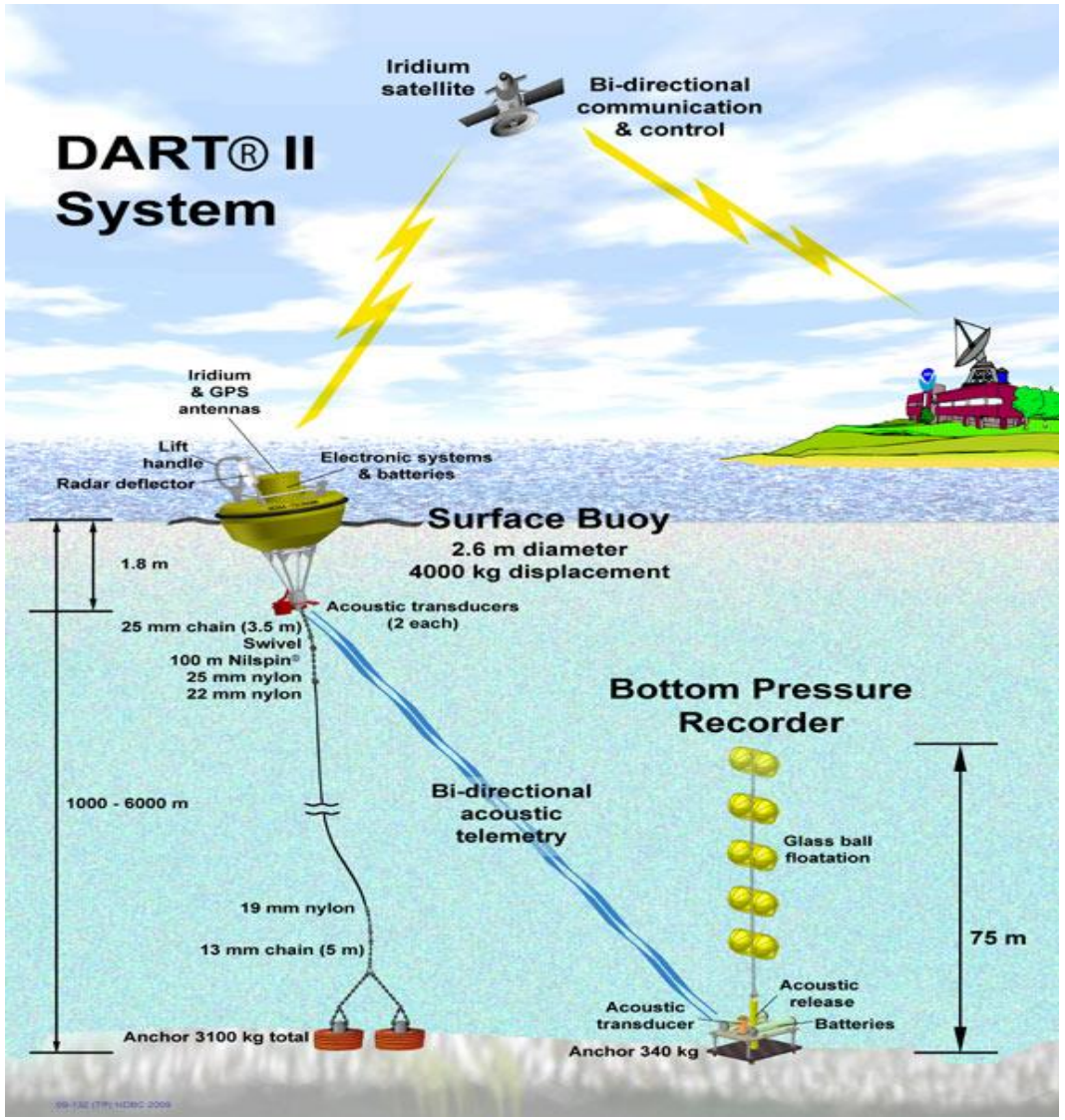
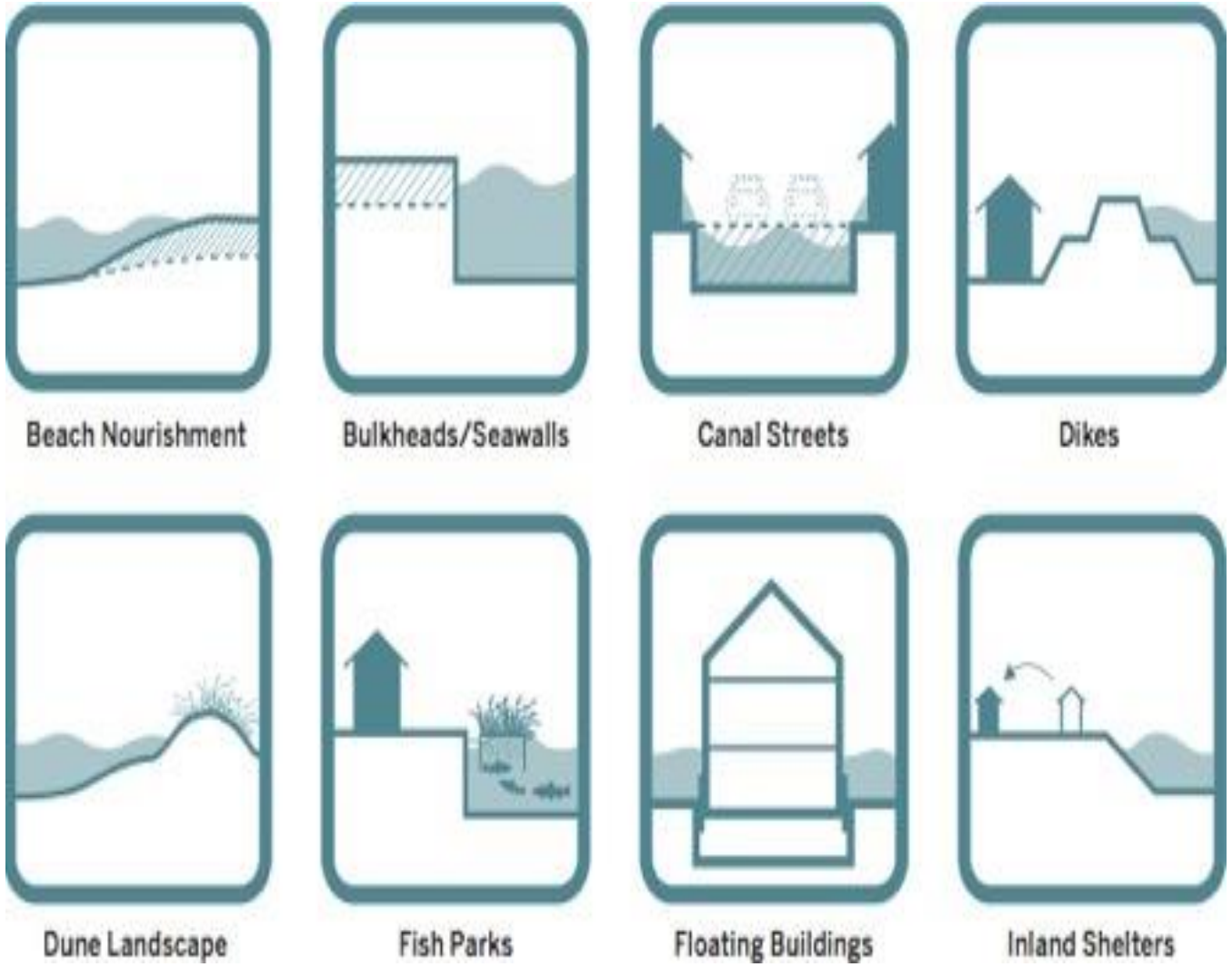


- **Seawalls, riprap, and breakwaters** constructed to protect buildings and roadways prevent coastal ecosystems from shifting landward naturally as water levels rise.
- Extensive shoreline development also inhibits coastal wetlands, salt **marshes and mangrove swamps from shifting landward in response to flooding water.**

- **Beach nourishment** by dumping sand from different places o shore. And then converting this into a dune landscape by growing vegetation over it.
- **Seawalls** construction of hard material, can be vertical, curved.
- **Dike** construction along the shore length.
- **Retreat:** Back stepping from the low lying area.
- **Accommodation :** Small measures taken, elevating buildings, emergency shelter making etc.



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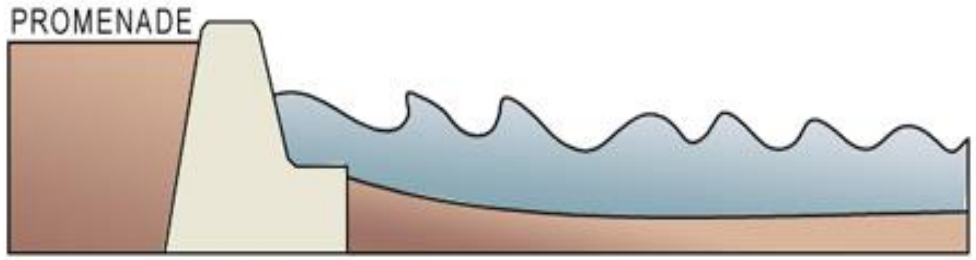
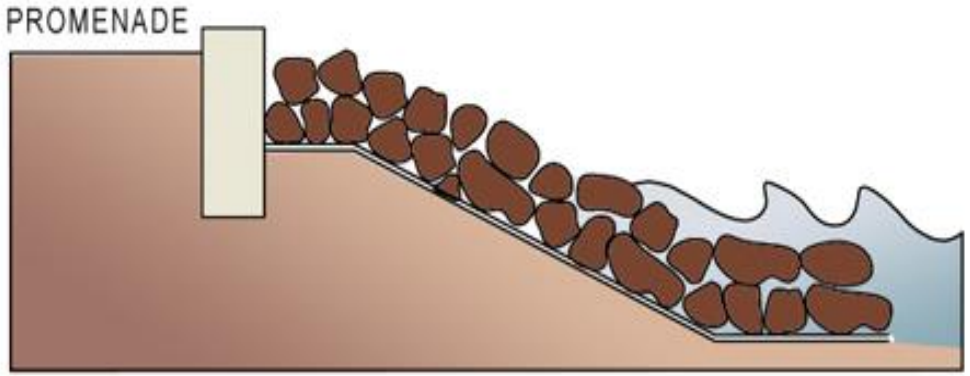
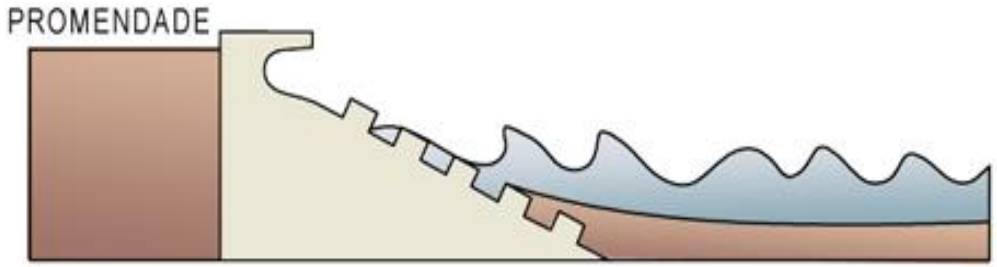
VERTICAL SEAWALL



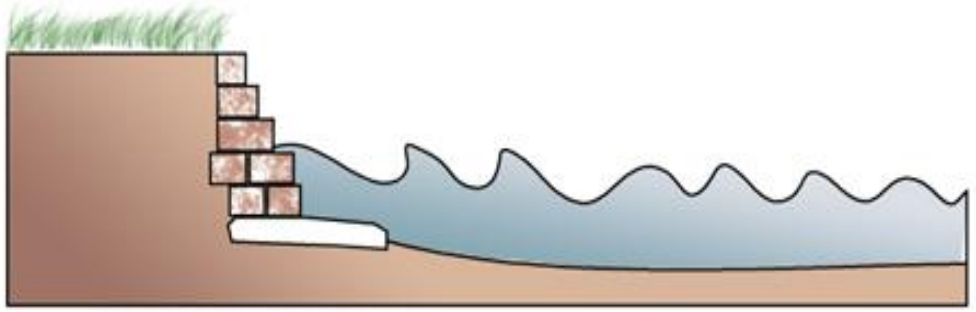
BOULDER MOUND



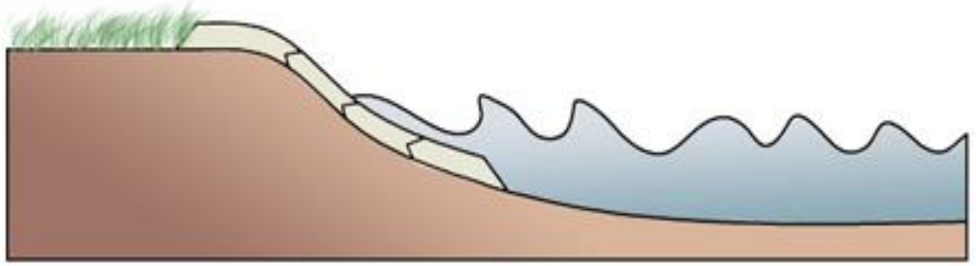
WALL WITH IRREGULAR FACE



GABION SEAWALL



CONCRETEINTERLOCKING BLOCKS



LANDSLIDES

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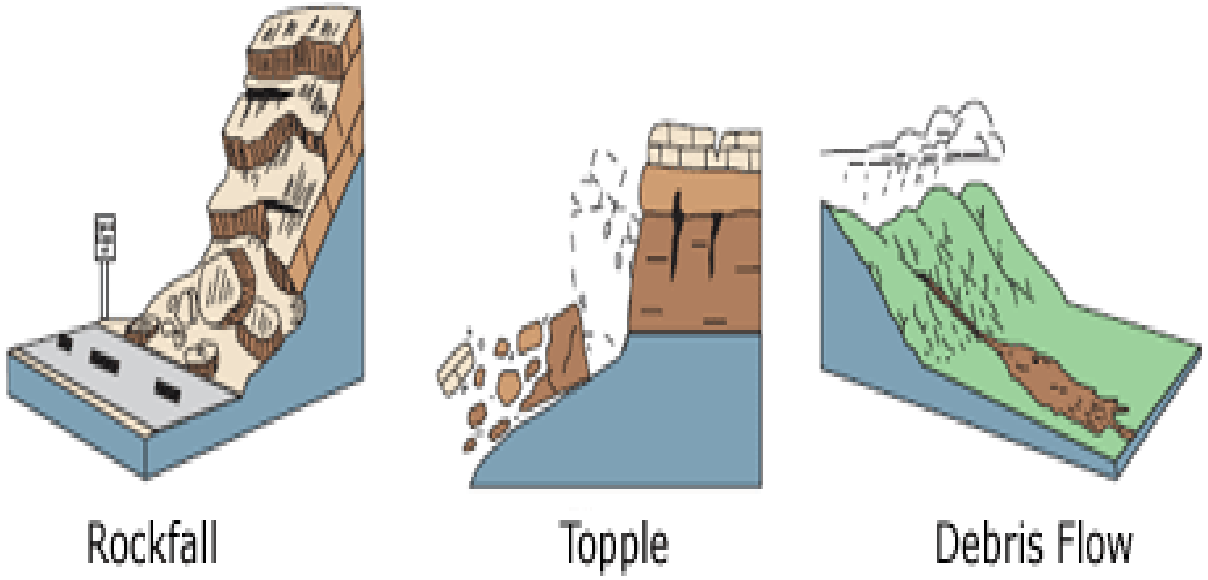
- A geological phenomenon which include wide range of ground movement such as rock falls, deep failure of slopes and shallow debris flow.
- PHASE 1: initial slope failure
- PHASE 2: subsequent transport
- PHASE 3 and the final deposition of the slide materials.

Natural causes of Landslides: Earthquakes, heavy rainfall, thunder and lightning may trigger failure of weak slopes, erosion by rivers and ocean waves create over steepened slopes

Human causes of Landslides: Deforestation and overgrazing, Mining: Poor construction, urbanization of hilly area, Vibrations from machinery, extensive agriculture practice, ground



Landslide Disaster Management

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Non structural Mitigation Measures

- Creation of awareness among local population.
- Insurance Mud flow policies from National Flood Insurance Program.
- Total avoidance of landslide hazard areas or by restricting, prohibiting, or imposing conditions on hazard-zone activity.
- Local governments can reduce landslide effects through land-use policies and regulations.
- By properly evaluating the hazard potential of a site, built or unbuilt.

Structural Measures:

- Stabilizing the slope
- Covering the landslide with an impermeable membrane
- Directing surface water away
- Draining ground water
- Minimizing surface irrigation.
- Proper land use measures
- Construction of retaining walls.
- Construction of channels or deflection walls

Preparedness/response
Forewarning before a landslide

- **Springs or saturated grounds**
- New cracks or bulges in the ground can predict landslides.
- If concrete foundations crack or move, a landslide may occur.
- After a landslide, flooding may occur.
- Evacuate people to areas of Safety from Landslides
- Areas that are safe from landslides are hard, non-jointed bedrock that has not moved in the past.

Landslide Recovery

- **Replant damaged ground to prevent flash flooding.**
- Compensation and Rehabilitation
- Repairs and construction of infrastructural facilities
- Relief to all affected people and relief for economic construction
- Camp-work and employment
- .